

Introduction + Goals

Have you ever had to wait over 10 minutes at a bus stop with no bench? What if it was raining and there was no shelter? Or you had a snack wrapper but there was no trashcan to discard it into? Lacking bus stop infrastructure can negatively affect all community members, even if only to the degree of fostering littering due to lack of trashcans. Not having benches or shelters can especially impact bus reliant commuters who cannot stand for long periods of time due to things like physical ailments, age, or pregnancy. This project aims to identify reasons why bus stops lack infrastructure in Montgomery County and suggest paths for improvement.

Datasets

The main data for this project is the Bus Stops dataset from Data Montgomery. It includes stops maintained by both Montgomery County and municipalities, and with bus service from both Ride On and WMATA buses. From this dataset I was mainly interested in the variables 'Total Benches,' 'Total Shelters,' and 'Total Trashcans.' My supplementary dataset sourced from Maryland.gov included all RideOn bus stops in Montgomery County. Key variables from this dataset were 'Stop Code' and 'Town.'

Tools + Pre-processing

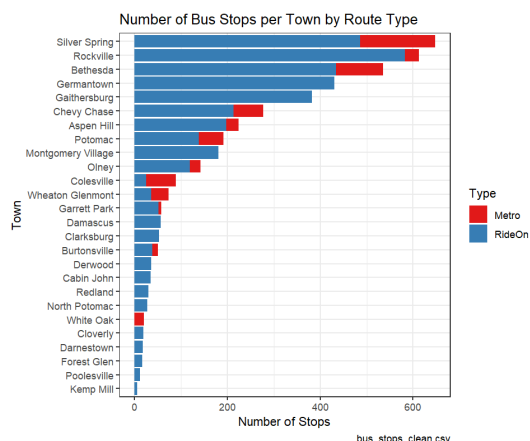
This project was completed entirely in R Studio utilizing various packages like tidyverse and stringr for data ingestion and cleaning, as well as sf for spatial nearest neighbor calculations and leaflet for interactive visualizations. My data processing stage consisted of standardizing variable names, removing commas from numeric entries, creating a new variable called 'Type' to distinguish between Metro and RideOn bus stops and joining the main and external datasets as mentioned previously.

Descriptive Statistics

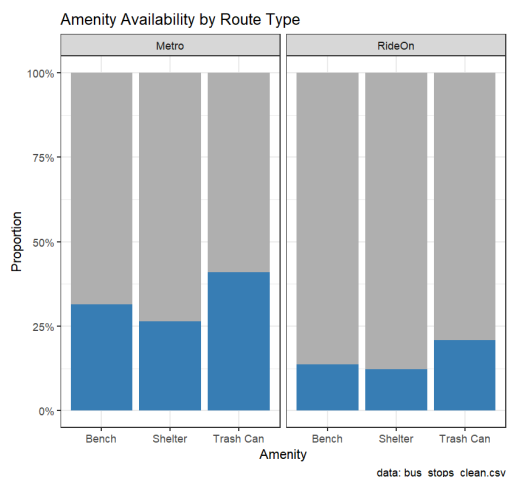
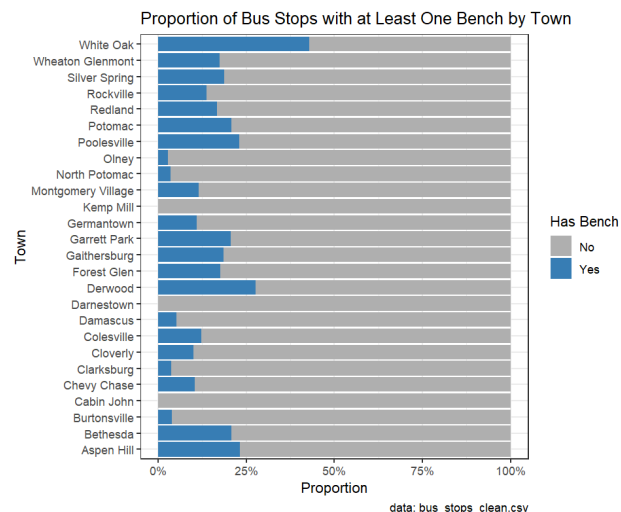
My final joined dataset represents 5253 bus stops in Montgomery County across 23 variables. There are 26 towns which include cities like Rockville and Gaithersburg as well as census-designated places which notably include populous places like Silver Spring, Bethesda and Germantown.

Visualizations and Products

The first visualization I'd like to introduce is a bar chart which gives an overview of the amount of bus stops in each town as well as how many of those stops only service RideOn routes or if they include Metro buses (fig. 1). It shows that Silver Spring has the most bus stops overall, closely followed by Rockville and Bethesda, which could suggest that population is closely tied to the number of bus stops in a given town. The chart also shows that the large majority of bus stops across towns only include RideOn routes. In this regard, however, White Oak stands out as it only has stops that include Metro routes. This inconsistency becomes interesting when looking at the next bar chart which shows the proportion of bus stops with at least one



bench for each town (fig. 2). This graph reveals that Whiteoak has the highest proportion of bus stops at a little less than 50%, while the majority of towns have proportions less than 25%. This finding made me wonder if the type of routes that a bus stop services, that being RideOn or Metro, had a significant effect on the amount of infrastructure, or amenities, a stop had. This led me to create my next visualization which shows the proportion of different amenities, those being benches, shelters, and trashcans, between RideOn and Metro bus stops (fig. 3). This graph showed that bus stops including Metro routes had higher percentages across all amenities, which would support the claim that there is an infrastructural difference between RideOn and Metro bus stops. To be sure that the



difference in proportions seen in the graph was statistically significant, I ran a chi-squared test which tested the same claim. This test resulted in all the amenities having significant p-values (< 0.05) even after Bonferroni correction. This means that the type of route that passes through a stop is strongly associated with whether a stop has amenities. While the chi-squared test doesn't tell us this directly, from the previous graph we know that this means that Metro bus stops generally have better infrastructure than RideOn bus stops.

The reasons for the gap in infrastructure between RideOn and Metro bus stops could be due to a couple of things. For starters, as seen in fig. 1, there are simply more RideOn only bus stops compared to stops that include Metro buses. This difference in frequency could exacerbate the gap in infrastructure between the two types. Another potential reason for this gap is funding. While both RideOn and Metro buses are funded by a combination of federal and state sources, a key difference between the two is that Metro buses charge a fare while RideOn buses have been free to all riders under 18 since 2019 and completely fare-free since June 29, 2025. Amidst this, Maryland is currently facing a transportation budget shortfall of \$2.1 billion, which can put stress on transit operating budgets (Goffman). It is also to be considered that the Metro operates across the DMV while RideOn routes operate mainly within the county. This means that unlike the RideOn system, Metro buses can benefit from funding across Maryland, Virginia and DC. Having a broader range of funding can place less stress on local funding sources which can allow those sources to be focused toward what will directly benefit the surrounding communities.

To summarize some key takeaways, this project found that Metro bus stops have significantly better infrastructure than RideOn bus stops. This could be because they are

included in a smaller ratio of the total bus stops across the county, their use of fares and/or their larger operating zone. The budget stress on the Maryland Department of Transportation could also impact the RideOn system's ability to develop better infrastructures of their bus stops.

Reflection

Something that I found challenging with this project was building a connected data story for the final product. Even within this report I found it difficult to tie in the Nearest Neighbors (NN) analysis and leaflet that I created during my main [analysis](#), which is why I opted to focus on the analysis regarding the type of bus stops (RideOn vs. Metro). Relating to the NN analysis, another thing I found challenging was implementing this nearest neighbors distance analysis as it was my first time working with the model in R Studio. However, I found it to be possible with the help of documentation resources and utilizing R Studio's built in Posit Assistant to troubleshoot code. On a different note, simply being able to complete this project was something that was rewarding to me. While there were many times that I wanted to give up, I was glad that I was able to put something out there. If I were to change something, I would want to add population and demographic data directly into my joined dataset. While I did consider these factors, specifically population, using separate sources, it could've been more meaningful if I included them as actual variables within the dataset like I originally planned to.

Next Steps and recommendations

To expand on this project, I would be interested to focus more on the nearest neighbors analysis and plot where bus stops are located in specific towns with large and small distances between bus stops with benches. Doing this could potentially give insight into what factors influence bench placement and where benches could be placed to lessen the distance between bus stops with benches. Another addition that could be meaningful to this project is creating a logistic linear regression with additional variables like population and demographic data to see the magnitude of impact different factors have on bus stop infrastructure. While the previously mentioned chi-squared test showed that type was significant in determining infrastructure, we don't know the size of that influence which a regression would express. Regarding recommendations, I would suggest focusing on towns like Olney and Clarksburg which have the lowest bench rates across towns. Additionally, considering the demographics of towns, such as a larger population of elderly people, could also be noteworthy in deciding where to build more infrastructure. While not specifically represented in this project's dataset, the town of Leisure World, located near Silver Spring, has a significant amount of active adult homeownership opportunities which cater to the needs of older adults (Montgomery Planning). This would make Leisure World a good candidate for bus stop infrastructure development as elderly people would significantly benefit from being able to sit down at bus stops.

Acknowledgements

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